Structured questions

|!|BCELC0030000001|!|

The electron micrographs below show some organelles that can be found in eukaryotic cells.

Electron micrograph A:

Organelle A (×15,000)



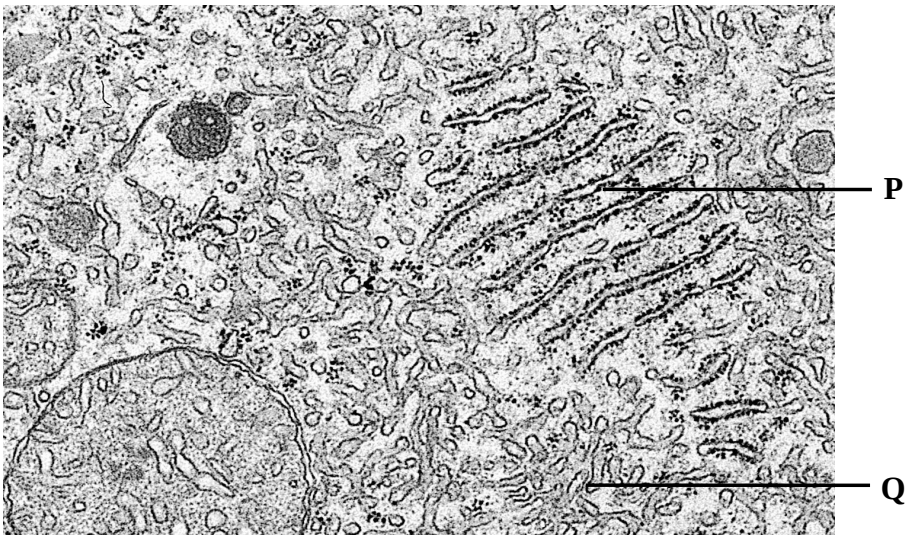
Electron micrograph B:

Organelle B (×72,000)



Electron micrograph C:

Endoplasmic reticulum (×16,000)



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(a) (i) What are the functions of organelles A and B? (2 marks)

(ii) Describe and explain how the features of organelles A and B facilitate their functioning. (4 marks)

(b) Structures P and Q have slightly different appearances.

(i) Describe a difference in appearance between structures P and Q as shown in Electron micrograph C. (1 mark)

(ii) Which structure, P or Q, is more abundant in enzyme-secreting cells? Explain your answer. (3 marks)

##

(a) (i) Organelle A contains chlorophyll which absorbs light energy for photosynthesis. (1)
Organelle B is the site of ATP production during aerobic respiration. (1)

(ii) The extensive network of thylakoid membranes in organelle A (1) provides a large surface area for the attachment of chlorophyll molecules for efficient light absorption. (1)
The inner membrane of organelle B is highly folded. (1)
This increases the surface area for the attachment of enzymes (or carriers) involved in respiration. (1)

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(b) (i) Structure P has ribosomes attached on it while structure Q does not. (1)

(ii) Structure P contains ribosomes which synthesize proteins. (1)

Enzymes are protein in nature. (1)

Therefore, a large amount of structure P is required in enzyme-secreting cells to synthesize enzymes. (1)

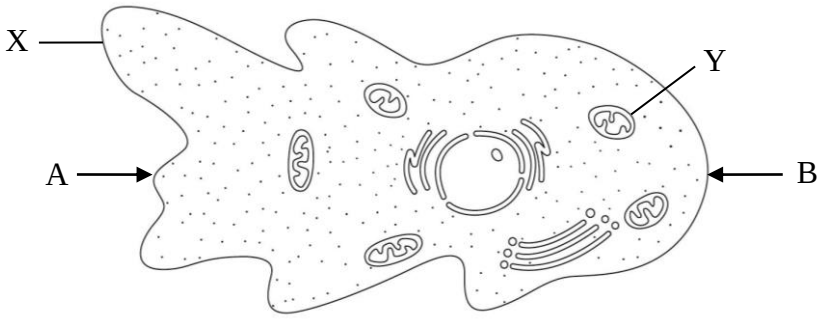
(10 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Applying concepts	4
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3

|!|BCELB0030000002|!| ★

The diagram below shows an *Amoeba*, a single-celled organism.



(a) Name structures X and Y. (2 marks)

(b) The magnification of the *Amoeba* in the drawing is 160x. Calculate the actual length between points A and B. Show your working and express your answer in μm (micrometre). (2 marks)

(c) Is *Amoeba* a prokaryote or eukaryote? Explain your answer. (3 marks)

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(d) The above *Amoeba* is drawn as seen under an electron microscope. Which structures cannot be seen when observed with a light microscope? (3 marks)

##

(a) X is the cell membrane. (1)
Y is the mitochondrion. (1)
(b) Actual length = 7.3 cm / 160 = 0.045625 cm x 10000 (1)
= 456.3 μm (1)
(c) *Amoeba* is a eukaryote. (1)
It has a membrane-bound nucleus and (1)
membrane-bound organelles. (1)
(d) Mitochondria / endoplasmic reticulum / ribosomes / nucleolus (Any ***three***, 3)

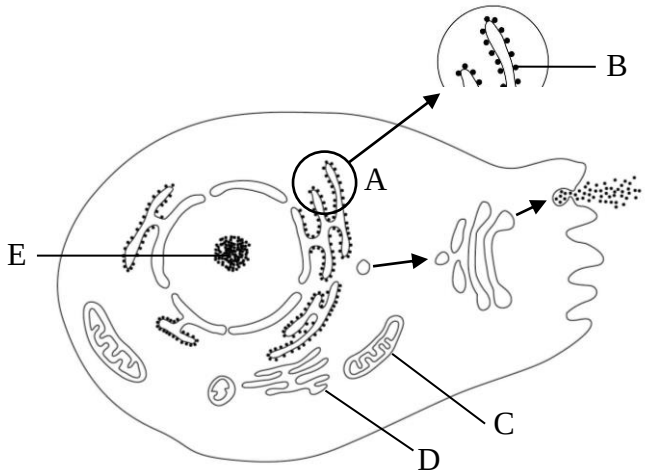
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Mathematical skills	2
(c)	Applying concepts	3
(d)	Understanding basic concepts	3

|!|BCELA0030000003|!| ★★

The diagram below shows a cell from the pancreas that secretes enzymes, and its sub-cellular structures.



(a) (i) Name structures A and D. (2 marks)

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(ii) Compare the similarity and difference in their structure. (2 marks)

(b) Structures A and B are closely related in functioning. Explain how. (2 marks)

(c) State one type of cells in the human body that have a large amount of structure C. (1 mark)

(d) Explain how this cell is adapted for the secretion of enzymes. (Hint: What is the nature of enzymes?) (1 mark)

##

(a) (i) A is rough endoplasmic reticulum. (1)
D is smooth endoplasmic reticulum. (1)
(ii) Both rough ER and smooth ER are networks of interconnected membrane-bound sacs. (1)
Rough ER has ribosomes attached on its surface, but smooth ER has no ribosomes attached. (1)
(b) Structure B is the site of protein synthesis. (1)
The proteins produced are transported by structure A. (1)
(c) Liver cells / muscle cells / sperm cells (1)
(d) This cell has a large amount of rough ER for the synthesis and transport of proteins (enzymes). (1)

(8 marks)

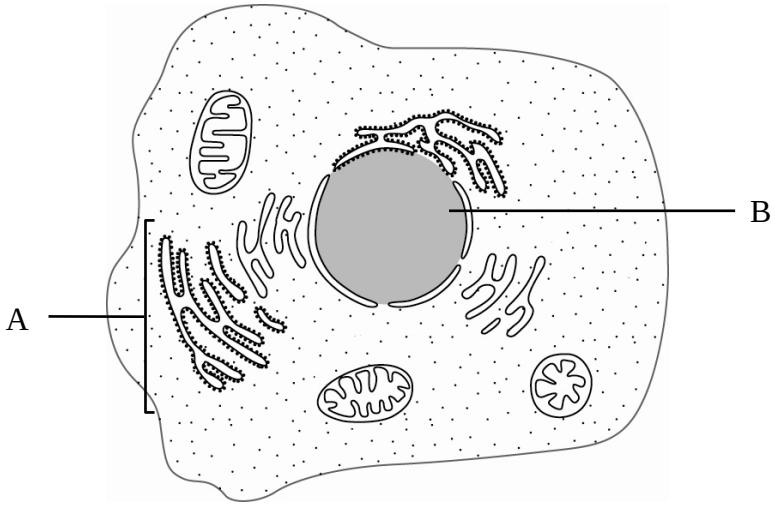
##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)(i)	Practical skills	1
(b)(ii)	Practical skills	1
(c)	Applying concepts	2

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|!|BCELA0030000004|!| ★★ **Cross topic**

The diagram below shows a human intestinal cell observed with an electron microscope. This cell actively secretes a type of digestive enzyme.



The table below shows the percentage by total cell volume and the number of different types of structures in this cell.

Structure	Percentage by volume (%)	Number
Lysosomes	1	300
Nucleus	4	1
Rough endoplasmic reticulum	10	1
Mitochondria	22	1700

(a) Describe the role of structures A and B in producing the digestive enzyme. (2 marks)

(b) Explain the evidence from the table which supports that this cell synthesizes large amounts of digestive enzymes. (3 marks)

(c) Explain the significance of a large number of mitochondria in the cell in relation to the function of the cell. (2 marks)

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(d) Which organelle, mitochondrion or lysosome, is larger? Support your answer with the given data. (3 marks)

##

(a) Structure A helps transport proteins produced at the ribosomes. (1)
Structure B carries genetic information / DNA which directs protein synthesis. (1)
(b) The percentage by volume of rough endoplasmic reticulum is high. (1)
The ribosomes attached on the surfaces of rough endoplasmic reticulum are responsible for protein synthesis. (1)
Large amounts of ribosome could synthesize large amounts of enzymes which are proteins. (1)
(c) Protein synthesis requires energy. (1)
Mitochondria release energy / produce ATP from respiration. (1)
(d) The percentage by volume of one mitochondrion = 22% / 1700 = 0.01% (1)
The percentage by volume of one lysosome = 1% / 300 = 0.003% (1)
The mitochondrion is larger than the lysosome. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	2
(c)	Applying concepts	2
(d)	Analysing data	3

|!|BCELB0030000005|!| ★ SI

A student cut a thin section of tissue from a potato and examined it with a light microscope.

(a) Explain what is meant by a tissue. (1 mark)

(b) Explain why the section of tissue to be viewed under a light microscope should be thin. (2 marks)

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(c) Potato cells contain starch. Describe how the student could find out where in the cells the starch was present. (3 marks)

(d) The cell walls of potato cells contain cellulose. Cellulose and starch are both carbohydrates. Describe ***two*** ways in which cellulose is similar to starch. (2 marks)

##

(a) A tissue consists of a group of similar cells working together to perform a specific function. (1)

(b) Light can pass through the tissue. (1)
Details of cells would not be obscured by too many cells underneath. (1)

(c) Add iodine solution to the potato tissue. (1)
Observe the potato tissue with a light microscope. (1)
The starch in the cells would be stained blue black. (1)

(d) Starch and cellulose are polymers of glucose molecules. (1)
Starch and cellulose are insoluble in water. (1)
(Or other reasonable answers)

(8 marks)

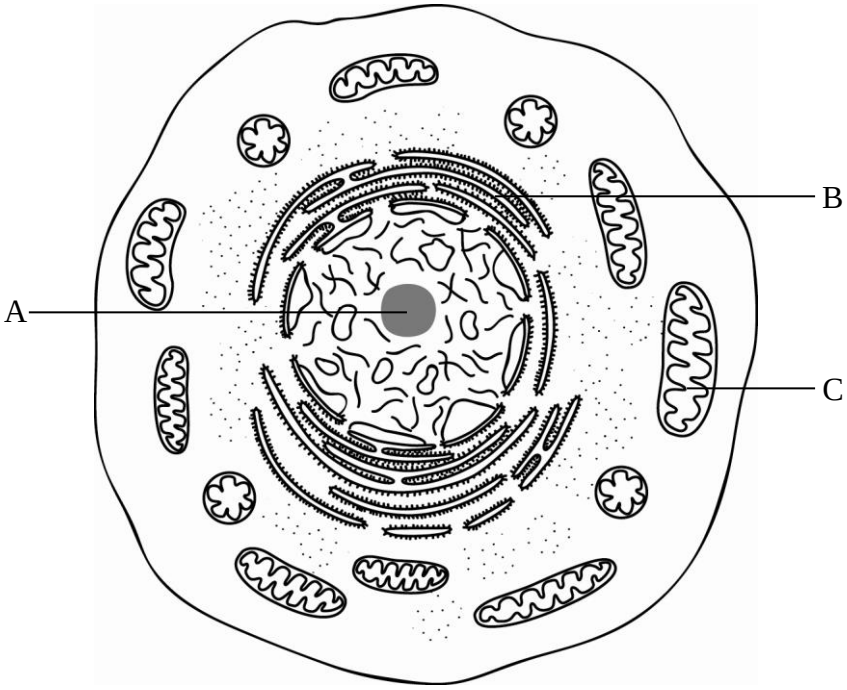
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Practical skills	2
	Applying concepts	1
(d)	Understanding basic concepts	2

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|!|BCELB0030000006|!| ★ **Cross topic**

The following diagram shows a type of white blood cell called B cell.



(a) Outline the roles of structures A and C in protein synthesis. (2 marks)

(b) mRNA is translated to form a polypeptide in structure B. Describe the process. (4 marks)

(c) (i) Name a protein synthesized by this cell. (1 mark)

(ii) State the function of this protein. (1 mark)

##

(a) The DNA in structure A contains the genetic information that codes for the amino acid sequence of proteins. (1)

Structure C supplies energy for protein synthesis (1)

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(b) mRNA attaches to the ribosome on structure B. (1)
tRNA molecules bring the corresponding amino acids to the ribosome. (1)
The anticodon of the tRNA binds with the complementary codon on the mRNA. (1)
Peptide bonds are formed between adjacent amino acids, forming a polypeptide chain that will form a protein. (1)

(c) (i) antibody (1)
(ii) to act against antigens / pathogens / toxins (1)

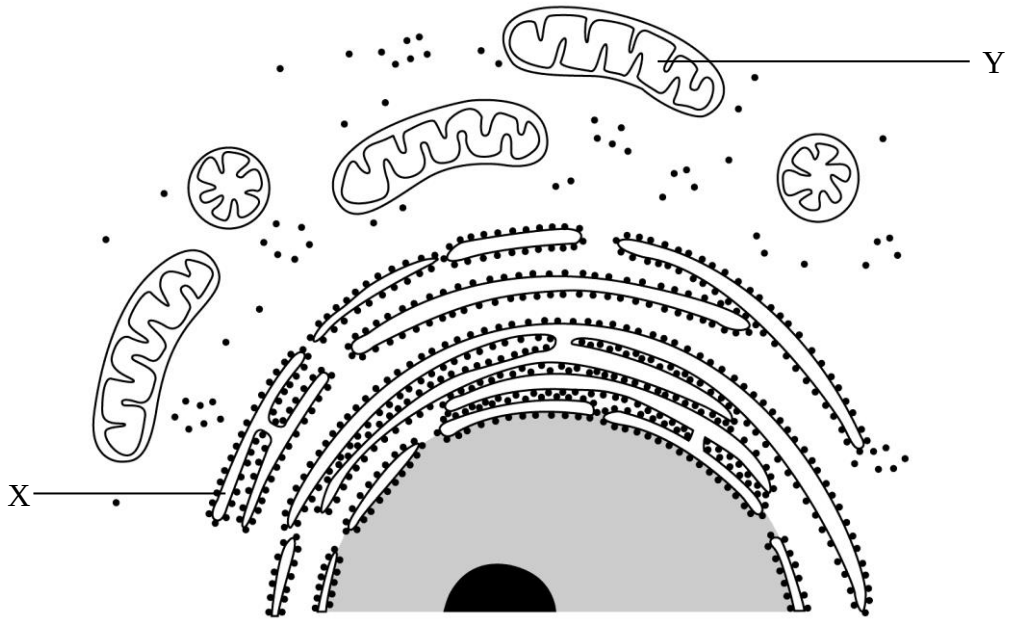
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	4
(c)(i)	Applying concepts	1
(c)(ii)	Understanding basic concepts	1

|!|BCELB0030000007|!| ★ **Cross topic**

The diagram below shows a part of an animal cell as seen under a microscope.



(a) State the type of microscope used to obtain the above view. (1 mark)

(b) Name the organelles labelled X and Y. (2 marks)

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(c) Describe how the structure of organelle X is adapted to its function. (3 marks)

(d) Explain why organelle Y, as shown in the diagram, has two different appearance. (2 marks)

(e) The epithelial cells of the ileum have abundant organelles Y. Explain how this is an adaptation for the function of these cells. (3 marks)

##

(a) transmission electron microscope (1)

(b) X: rough endoplasmic reticulum (1)
Y: mitochondrion (1)

(c) Organelle X (rough endoplasmic reticulum) is a network of interconnected membrane-bound sacs, (1)
which increases the surface area for accommodating more ribosomes. (1)
More proteins can then be synthesized from the ribosomes. (1)

(d) Mitochondria are three-dimensional structures. (1)
Different mitochondria may be in different orientations. (1)
OR
Some are transverse sections (1)
and some are longitudinal sections. (1)

(e) The absorption of digested food substances (e.g. glucose, amino acids) through the epithelium of the villi into the blood capillaries (1)
relies on active transport (1)
Active transport requires lots of energy / ATP formed from the abundant mitochondria. (1)

(11 marks)

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##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Understanding basic concepts	1
	Applying concepts	2
(d)	Applying concepts	2
(e)	Applying concepts	3

|!|BCELB0030000008|!| ★

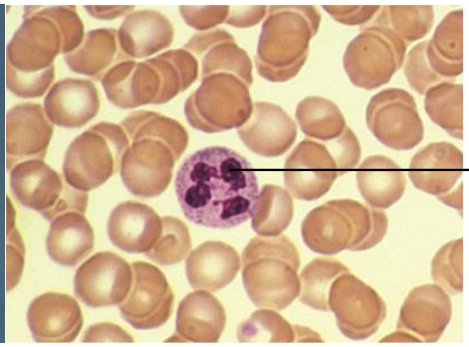
The photographs below show four types of cells.

Cell A



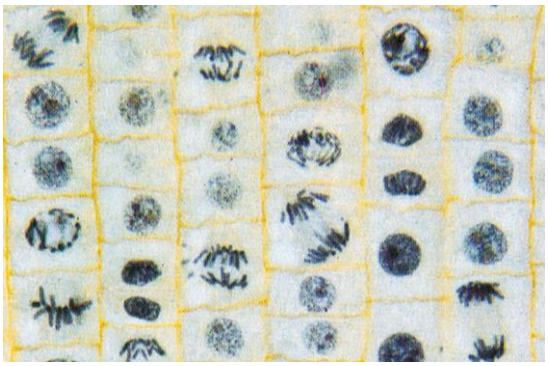
(x150)

Cell B



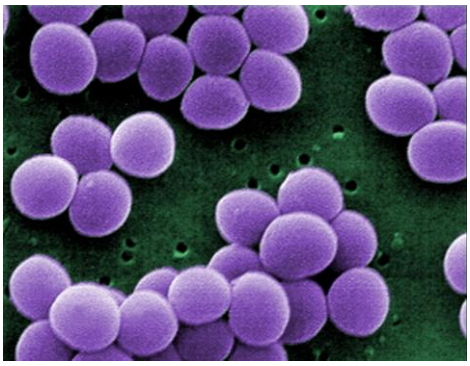
(x1000)

Cells C



(x400)

Cells D



(x20 000)

(a) (i) Classify cells A, B, C and D into prokaryotic cells and eukaryotic cells. (2 marks)

(ii) State ***two*** differences between prokaryotic cells and eukaryotic cells. (2 marks)

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(b) Which cells are photographed under an electron microscope? Give evidence to support your answer. (2 marks)

(c) Name structure X and state ***two*** of its functions. (3 marks)

##

(a) (i) Prokaryotic cells: Cells D (1)

Eukaryotic cells: Cells A, B and C (1)

(ii) Any ***two*** of the following comparisons: (2)

	Prokaryotic cells	**Eukaryotic cells**
•	usually smaller in size	usually larger in size
•	without nucleus	with a membrane-bound nucleus
•	its DNA lying free in the cytoplasm	its DNA enclosed in the nuclear membrane
•	without membrane-bound organelles	with membrane-bound organelles
•	some with cell wall which is not made of cellulose	some with cell wall made of cellulose

(b) Cells D (1)

They appear to be three-dimensional which can be observed under a scanning electron microscope. (1)

(c) X is the nucleus. (1)

It contains the genetic material DNA. (1)

It controls all cellular activities. (1)

(9 marks)

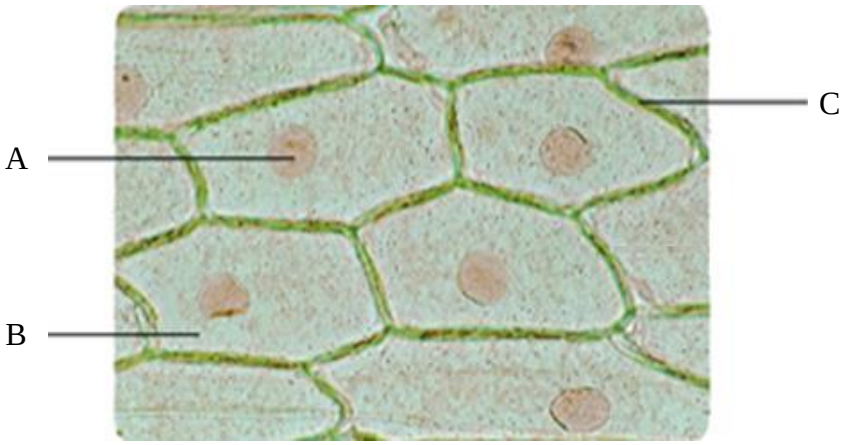
##

	Skills assessed	**Marks**
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Analysing data	1
	Understanding basic concepts	2

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|!|BCELB0030000009|!| ★ **Cross topic**

The following photomicrograph shows some onion epidermal cells.



(a) Name a type of stain that is used to make the onion epidermal cells become easily observable. (1 mark)

(b) Which labelled structure is absent in a white blood cell? (1 mark)

(c) Other than the labelled structures, name a cell structure that can be found in leaf mesophyll cells but are absent in onion epidermal cells. (1 mark)

(d) Compare structure C and the cell membrane in terms of

(i) chemical nature. (2 marks)

(ii) the mechanism by which water-soluble substances pass through them. (2 marks)

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(e) With reference to the photomicrograph, draw a labelled high-power diagram of the onion epidermal cells in the space below. (4 marks)

##

(a) iodine solution (1)

(b) C (cell wall) (1)

(c) chloroplast (1)

(d) (i) C is composed of cellulose and (1)
the cell membrane is composed of phospholipids and proteins. (1)

(ii) Water-soluble substances can pass through C freely by diffusion, while water-soluble substances pass through the cell membrane only through some channel proteins or carrier proteins. (1) (1)

(e) Correct title (1)
Resemblance of drawing (1)
Correct labels (any ***two***, 1 mark each) (2)

(11 marks)

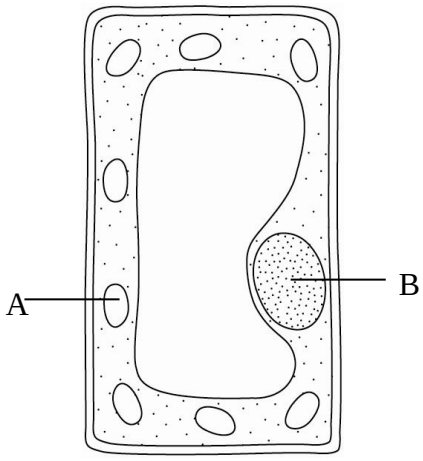
##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Analysing data	1
(c)	Understanding basic concepts	1
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2
(e)	Presenting information	4

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|!|BCELC0030000010|!|

The diagram below shows a mesophyll cell.



(a) Name and state ***one*** function of structures A and B. (4 marks)

(b) With reference to ***two*** features shown in the diagram, explain why the above cell is classified as a plant cell but ***not*** an animal cell. (2 marks)

(c) State ***two*** differences between this mesophyll cell and a root hair cell. (2 marks)

##

(a) Structure A: chloroplast (1)
Structure A is the site of photosynthesis which synthesizes food. (1)
Structure B: nucleus (1)
Structure B controls all activities of the cell. (1)

(b) Any ***two*** of the following: (2)

- The cell has a cell wall; while an animal cell has no cell wall.

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- There is a large central vacuole; while an animal cell has no vacuole or only a few small vacuoles.
- The cell has chloroplasts; while an animal cell has no chloroplast.

(c) Root hair cells have long extensions from the cells. (1)

Root hair cells have no chloroplasts. (1)

(Or other reasonable answers)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Understanding basic concepts	2
(b)	Analysing data	2
(c)	Understanding basic concepts	2

|!|BCELB0030000011|!| ★

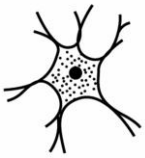
The following diagrams show four types of cells.



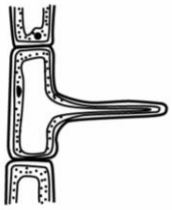
W



X



Y



Z

(a) (i) Using the given letters, indicate which are animal cells and which are plant cells. (2 marks)

(ii) According to the diagrams, state ***one*** feature that is found in all plant cells but not in animal cells. (1 mark)

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(b) Describe how cells X and Z are adapted to their functions. (4 marks)

##

(a) (i) Animal cells: W and Y (1)
Plant cells: X and Z (1)
(ii) Plant cells have cell wall and animal cells do not have cell wall. (1)
(b) Cell X has thick lignified cell wall which (1)
provides support to the plant. (1)
OR
Cells X have no cytoplasm and nucleus. They are joined end to end to form a continuous tube (1)
for the transport of water and minerals. (1)
Cell Z develops extensions / root hairs (1)
which increase the surface area for the absorption of water and minerals. (1)

(7 marks)

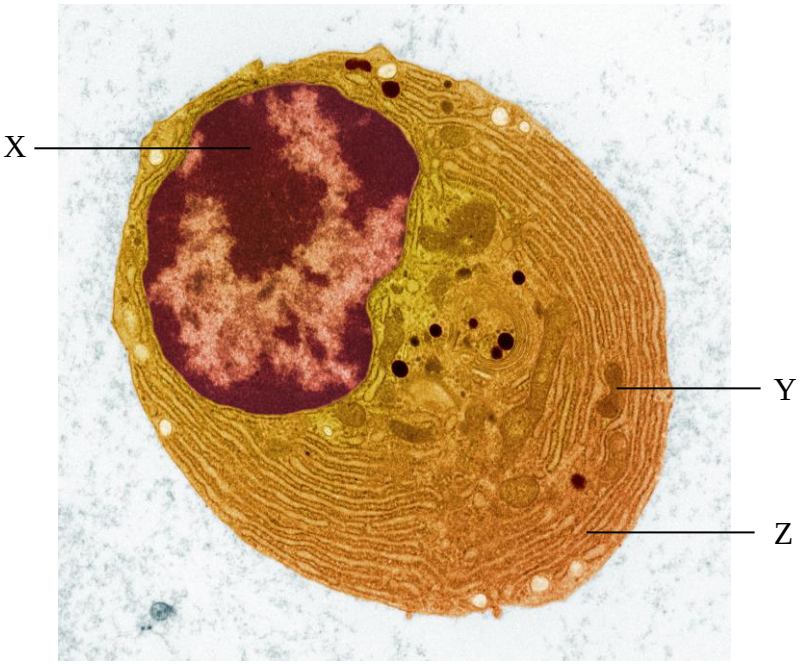
##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Analysing data	1
(b)	Analysing data	2
	Applying concepts	2

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|!|BCELB0030000012|!| ★

The following micrograph shows a plasma cell, which is an antibody-producing white blood cell.



(a) According to the micrograph, list ***two*** features showing that the plasma cell is an animal cell but not a plant cell. (2 marks)

(b) Structure X is found in the nucleus of the cell.
Identify structure X and state its function. (2 marks)

(c) Describe how structure Y is adapted to its function. (3 marks)

(d) In the plasma cell, there is a high density of structure Z.
(i) What is structure Z? (1 mark)

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(ii) State the significance of the high density of structure Z in the plasma cell.

(3 marks)

##

(a) Any ***two*** of the following: (2)

- No large vacuoles can be observed.
- No chloroplasts can be found in the cell.
- No cell wall can be found on the outside of the cell membrane.

(Or other reasonable answers)

(b) Nucleolus (1)

It is involved in making ribosomes. (1)

(c) Its inner membrane is highly folded to form numerous finger-like projections. (1)

This increases the surface area for accommodating more enzymes (1)

that help in ATP synthesis. (1)

(d) (i) rough endoplasmic reticulum (1)

(ii) Antibodies are proteins. (1)

The high density of structure Z can help in synthesizing more antibodies (1)

and help in transporting the antibodies for secretion. (1)

(11 marks)

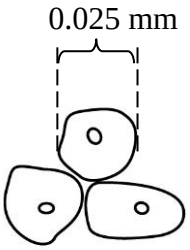
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Understanding basic concepts	3
(d)(i)	Analysing data	1
(d)(ii)	Applying concepts	3

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|!|BCELB0030000013|!| ★

Tom prepared a temporary mount of ox eye corneal cells. He observed the cells with a light microscope and made a drawing of the cells as shown below.



(a) The actual width of the cell between the two dotted lines is 0.025 mm. Calculate the magnification of the cells in the drawing. Show your working. (2 marks)

(b) Describe how the image observed with the microscope differs from the cells on the slide. (1 mark)

(c) State the change in each of the following if a higher magnification is used to observe the specimen.

(i) Area of the specimen observed (1 mark)

(ii) Brightness of the image (1 mark)

(iii) Details of the image (1 mark)

(d) State in which direction the image will move if Tom moves the slide to the left. (1 mark)

##

(a) Magnification =10 / 0.025 (1)
= 400X (1)

(b) The image is inverted upside down and reversed laterally. (1)

(c) (i) Area of the specimen observed becomes smaller. (1)

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(ii) The image becomes dimmer. (1)

(iii) More details of the image can be observed. (1)

(d) The image moves to the right. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(c)(iii)	Understanding basic concepts	1
(d)	Applying concepts	1

|!|BCELB0030000014|!| ★ SI

Read the passage below and answer the questions that follow.

Microscope

Robert Hooke developed the first microscope and used the term 'cells' to describe the tiny box-like units after he had observed the slices of cork from the bark of a tree in 1665. Later, during the period 1839–1858, Schleiden, Schwann and Virchow developed the cell theory. In 1933, a scientist called Ruska invented the electron microscope which can be used for medical and biological applications.

(a) State ***one*** structural difference between a cork cell and a human cell. (1 mark)

(b) State the ***three*** ideas of the cell theory. (3 marks)

(c) Describe how the development of microscopes has helped the study of different pathogens. (3 marks)

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(d) State the purpose of the following steps when preparing a temporary mount of onion epidermis for observation under a light microscope.

(i) adding iodine solution onto the epidermis (1 mark)

(ii) covering the epidermis with a cover slip (1 mark)

##

(a) A cork cell has a cell wall and a human cell has not. (1)

(b) All living organisms are made up of one or more cells. (1)

The cell is the basic unit of life. (1)

Cells arise from pre-existing cells. (1)

(c) Early invention of light microscopes allowed scientists to observe microscopic parasitic worms. (1)

Improvement of light microscopes with lens of higher magnification allowed scientists to observe bacteria of various sizes. (1)

The development of the electron microscope allowed scientists to observe different viruses. (1)

(d) (i) To make the transparent epidermal cells easier to be observed (1)

(ii) To flatten the epidermis / prevent the epidermis from drying up (1)

(9 marks)

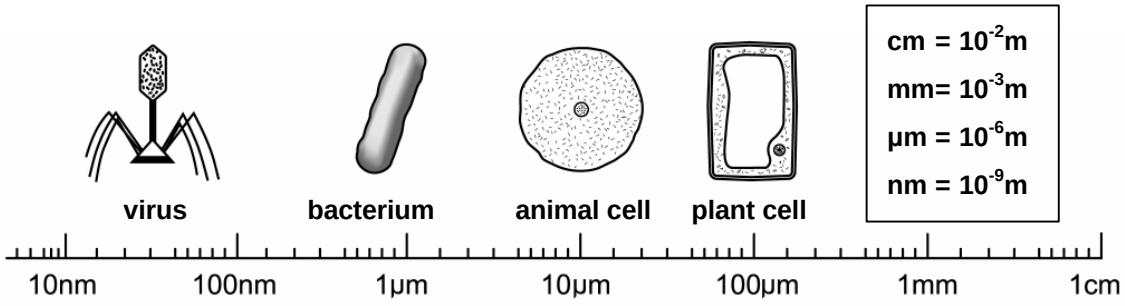
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Applying concepts	3
(d)(i)	Practical skills	1
(d)(ii)	Practical skills	1

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|!|BCELC0030000015|!|

The diagram below shows the relative sizes of a virus and some cells.



(a) Bacteria are prokaryotes. Describe how a bacterium differs from eukaryotic cells. (3 marks)

(b) How does a typical plant cell differ from a typical animal cell? (3 marks)

(c) Which of the four structures shown in the diagram cannot be seen under a light microscope, but can be seen under an electron microscope? (1 mark)

(d) State ***one*** advantage and ***one*** limitation of using the transmission electron microscope (TEM) for observing specimens. (2 marks)

##

(a) A bacterium does not have a membrane-bound nucleus as in eukaryotic cells / The genetic material of a bacterium lies free in the cytoplasm. (1)
A bacterium does not contain membrane-bound organelles as in eukaryotic cells. (1)
The size of a bacterium is much smaller than that of a eukaryotic cell. (1)

(b) A typical plant cell contains chloroplasts, while a typical animal cell does not contain chloroplasts. (1)
All plant cells have a cell wall, but animal cells do not have a cell wall. (1)

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Most plant cells have a large central vacuole, while animal cells have no or a few small vacuoles only. (1)

(c) Virus (1)

(d) Any ***one*** of the advantage: (1)

- Very small specimens can be observed with TEM.
- TEM has a high resolution.
- Internal structures of the specimens can be observed with TEM.

(Or other reasonable answers)

Any ***one*** of the limitations: (1)

- Living specimens cannot be observed with TEM.
- Very thin sections of the specimens must be prepared for TEM observation.

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	3
(c)	Analysing data	1
(d)	Understanding basic concepts	2

|!|BCELA0030000016|!| ★★ SI

Peter used a compound microscope to observe the longitudinal section of the skin epidermis. At first, he viewed the specimen as shown in **Figure I**. After he adjusted the microscope, he viewed the specimen as shown in **Figure II**.

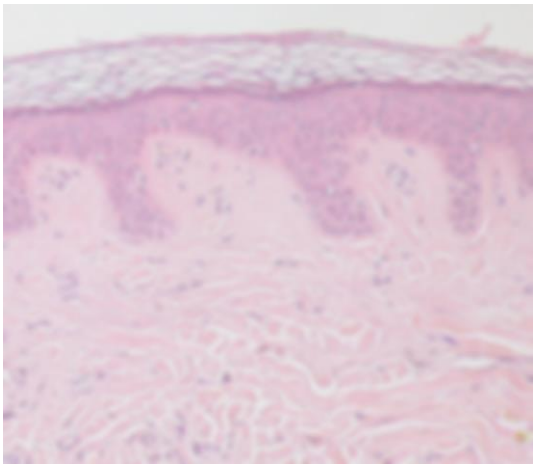


Figure I

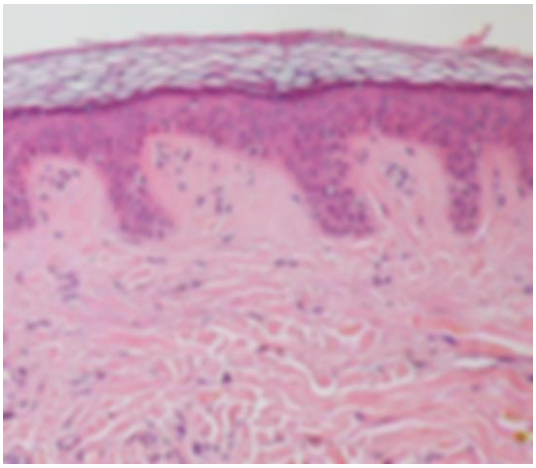


Figure II

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(a) Suggest how Peter could adjust the microscope to correct the problem shown in **Figure I**. (2 marks)

(b) The image formed as in **Figure II** is not in focus. Which part(s) of the microscope should be adjusted to obtain a sharp image? (1 mark)

(c) In order to get a more detailed image of the specimen, the magnification was changed from low power to high power.

(i) Write down ***two*** precautions when switching from low power magnification to high power magnification. (2 marks)

(ii) Compare the images observed at low power and high power magnifications. (2 marks)

(d) Explain why the skin is regarded as an organ. (1 mark)

##

(a) He could adjust the condenser, the iris diaphragm and the angle of the mirror (1)
to reduce the amount of light entering the condenser. (1)
Hence, the image would not be too bright.

(b) Slowly turn the fine adjustment knob. (1)

(c) (i) Look from the side when turning the nosepiece to select the high power objective so as to make sure the objective would not touch the slide or the stage. (1)

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Focusing should only be done by using the fine adjustment knob. (1)

(ii) Any ***two*** of the following comparisons: (2)

	Low power magnification	High power magnification
Area of the specimen observed	larger	smaller
Details of the image	fewer	more
Brightness of the image	brighter	dimmer

(d) The skin is composed of several types of tissues working together to perform functions such as body defence and body temperature regulation. (1)

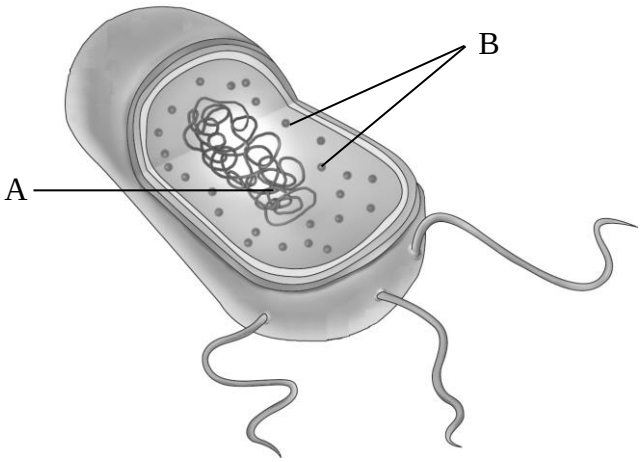
(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2
(d)	Applying concepts	1

|!|BCELC0030000017|!|

The following diagram shows a prokaryotic cell.



(a) Name structures A and B. (2 marks)

(b) State ***one*** function of structures A and B. (2 marks)

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(c) There are many structural differences between a prokaryotic cell and a eukaryotic cell.

(i) State ***two*** structures that are present in a eukaryotic cell but absent in a prokaryotic cell. (2 marks)

(ii) State ***one*** structure that is present in the cytoplasm of a prokaryotic cell but is absent in a eukaryotic cell. (1 mark)

##

(a) A: DNA (1)
B: ribosomes (1)

(b) A: stores genetic information (1)
B: site of protein synthesis (1)

(c) (i) Any ***two*** of the following: (2)
- true nucleus
- mitochondria
- (rough) endoplasmic reticulum

(Or other reasonable answers)

(ii) plasmid / coiled loop of DNA (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	1

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